

Prasun Rimal

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EDUCATION

St. Joseph's University New York

Bachelor of Science in Mathematics/Computer Science

GPA: 3.811/4.000

Honors: Sigma Iota Chi, Delta Epsilon Sigma, Dean's List: 6 semesters

Selected Coursework: Data Structures, Algorithm Design and Analysis, Software Engineering, Database Systems, Operating Systems, Advanced Computer Programming, Data Science, Probability and Statistics, Linear Algebra

May 2026

Brooklyn, NY

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, R, C++, MATLAB, HTML, CSS

Backend: Spring Boot, Spring Security, Spring Data JPA, JWT, Flask, Django, Express, REST APIs

Frontend: React, Vite, Tailwind CSS, Recharts, Leaflet, JavaFX

Data/ML: R/tidyverse, Pandas, NumPy, ggplot2, logistic regression, statistical testing, data visualization

Databases/Cloud: PostgreSQL, MySQL, MongoDB, Oracle, Neon, Render, Vercel, Docker, AWS

Core Concepts: Data structures, algorithms, graph algorithms, optimization, database design, mathematical modeling

PROJECTS

MockMarket: Full-Stack Paper Trading Platform

[Live Demo](#) — [GitHub](#)

Java 21, Spring Boot, React, TypeScript, PostgreSQL, JWT, Finnhub API, Vercel, Render, Neon

- Built and deployed a full-stack paper trading platform that gives each registered user a simulated \$100,000 account for buying, selling, and tracking stocks with live market quotes.
- Developed a Spring Boot backend with JWT authentication, BCrypt password hashing, protected REST APIs, and PostgreSQL persistence for users, trades, watchlists, accounts, and holdings.
- Implemented precise financial calculations with **BigDecimal**, including cash balance updates, average cost tracking, holdings value, portfolio value, and gain/loss calculations.
- Integrated Finnhub market data for live quotes and company search, including watchlist lookup by ticker or company name and visible refresh controls across core dashboard views.
- Designed a polished React/TypeScript dashboard with portfolio analytics, trade history, holdings sell controls, watchlist management, and responsive trading-terminal styling.

Manhattan Last-Mile Delivery Optimizer: SURF Research Project

[GitHub](#)

Python, Flask, NetworkX, Pandas, Leaflet, OpenStreetMap

- Built the technical artifact for a Summer Undergraduate Research Fellowship project focused on last-mile delivery optimization in Manhattan.
- Implemented Dijkstra's algorithm and A* search to compare shortest-distance routes against delivery-aware routes in a graph model enriched with traffic, curb-access, parking/loading, and truck-route constraints.
- Designed a multi-objective cost function that balances distance, congestion, curb limitations, and route legality to better reflect real urban delivery conditions.
- Developed an interactive Flask/Leaflet web app that visualizes baseline and optimized routes with route metrics, data coverage indicators, and algorithm agreement checks.
- Demonstrated an optimized sample route that added only 35.01 meters while reducing modeled delivery cost by 34.62 units.

NYC Motor Vehicle Collision Analysis

[GitHub](#)

R, tidyverse, ggplot2, statistical testing, logistic regression, spatial analysis

- Analyzed 187,082 NYC motor vehicle collision records from January 2024 to February 2026 to identify crash patterns, injury risk factors, and predictive signals.
- Built a reproducible R workflow for cleaning, feature engineering, borough-level analysis, time-of-day trends, injury/fatality rates, vehicle-type comparisons, and spatial coordinate coverage.
- Used chi-square tests to evaluate associations between injury outcomes and borough, contributing factor, and vehicle type.
- Trained and evaluated a logistic regression model for injury prediction, improving accuracy from about 55.5% to 63.16% with an AUC of 0.673.

- Produced publication-ready tables and visualizations for crash counts, injury rates, hourly trends, severity distribution, vehicle risk, ROC analysis, and spatial crash density.

EXPERIENCE

Summer Research Fellow, Summer Undergraduate Research Fellowship (SURF) June 2025 – August 2025
St. Joseph's University New York *Brooklyn, NY*

- Conducted faculty-advised undergraduate research on graph-based optimization strategies for last-mile delivery under real-world urban constraints.
- Met weekly with a research advisor to review progress, refine methodology, troubleshoot modeling decisions, and strengthen the direction of the project.
- Wrote a formal research paper documenting the problem motivation, data sources, algorithmic approach, optimization framework, findings, and limitations.
- Presented research findings to an audience, explaining the routing model, delivery-aware cost function, visual results, and trade-offs between baseline and optimized routes.

STEM Instructor Intern June 2025 – August 2025
Lavner Education *Pleasantville, NY*

- Taught STEM-focused camps for students ages 8–14 in web development, game design, introductory programming, and technical problem solving.
- Guided students through coding projects, debugging tasks, and hands-on activities while adapting explanations to different skill levels.
- Strengthened communication and leadership skills by breaking down programming concepts into clear, age-appropriate lessons.

LEADERSHIP

Vice President, South Asian Student Association August 2025 – May 2026
St. Joseph's University New York *Brooklyn, NY*

- Helped lead a campus cultural organization focused on community building, student representation, and South Asian cultural programming.
- Coordinated with executive board members to plan events, organize meetings, promote student engagement, and support inclusive programming for the broader campus community.
- Contributed to event logistics, outreach, and communication efforts for cultural celebrations, student activities, and organization-led initiatives.

AWARDS AND HONORS

- Sigma Iota Chi Honor Society
- Delta Epsilon Sigma Honor Society
- Dean's Scholarship Recipient
- Dean's List, 6 semesters